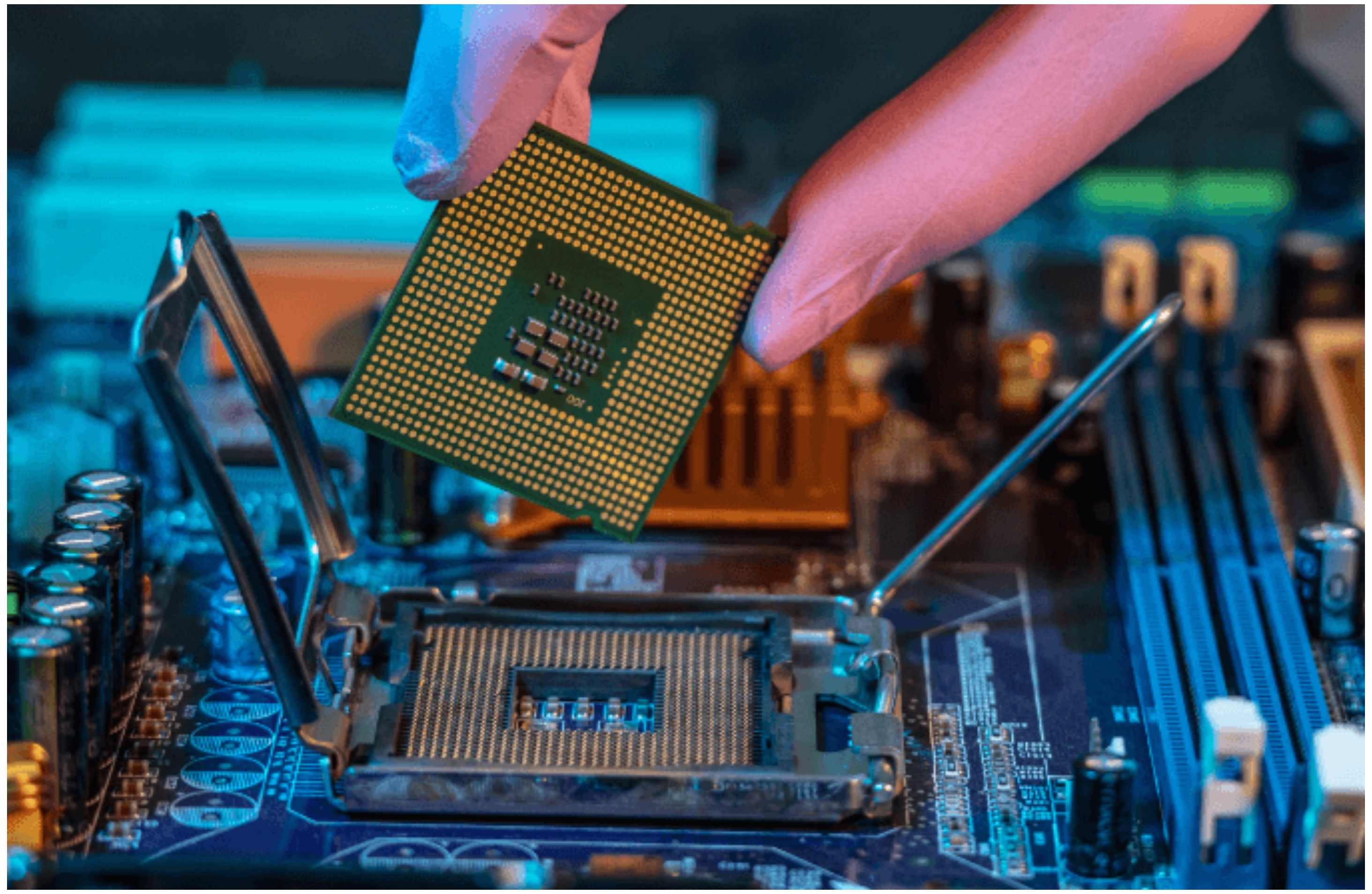
Enhancing Student Self-Efficacy and Interest in Microelectronics through Immersive Virtual Reality in an Informal Learning Environment

[1]Yupei Duan, [1]Xinhao Xu, [2]Fang Wang, [3]Chi-Ren Shyu, [3]Syed Kamrul Islam, [3]Sazia Eliza, [4]Jim Flink, [1]Yuanyuan Gu, [5]Hao He, [1]Shangman Li, [2]Scottie Murrell, [2]Amith Nalmas, and [3]Mahmoud Almasri

{[1]School of Information Science & Learning Technologies, [2]Department of Engineering & Information Technology, [3]Department of Electrical Engineering & Computer Science, [4]School of Journalism}, University of Missouri, Columbia, MO, United States
[5]School of Library & Information Management, Emporia State University, Emporia, KS, United States

Introduction

- Microelectronics powers AI, IoT, and advanced computing systems
- The semiconductor industry faces critical workforce shortages
- STEM learners lack hands-on exposure and practical training opportunities
- Virtual Reality provides immersive, scalable environments to bridge theory and practice



Problem Statement

- Hands-on lab experiences in microelectronics are expensive and limited
- STEM interest and self-efficacy decline over time (leaky pipeline)
- Students lack early exposure and practical engagement

Grant #1946619

Microelectronics Bootcamp

- AN NSF-supported 1.5-day microelectronics bootcamp (Fall 2024)
- Targeted undergraduate students with diverse academic backgrounds
- Activities included:
  - Lectures on semiconductor fundamentals
  - Career discussions with industry guests
  - Lab visiting;
  - **VR-based hands-on training using iVRLab**

iVRLab Project Overview

- 1-hour hands-on VR session with facilitators
- Pre/post surveys and follow-up interviews
- Single-user VR experience with scaffolded tutorial

(a)

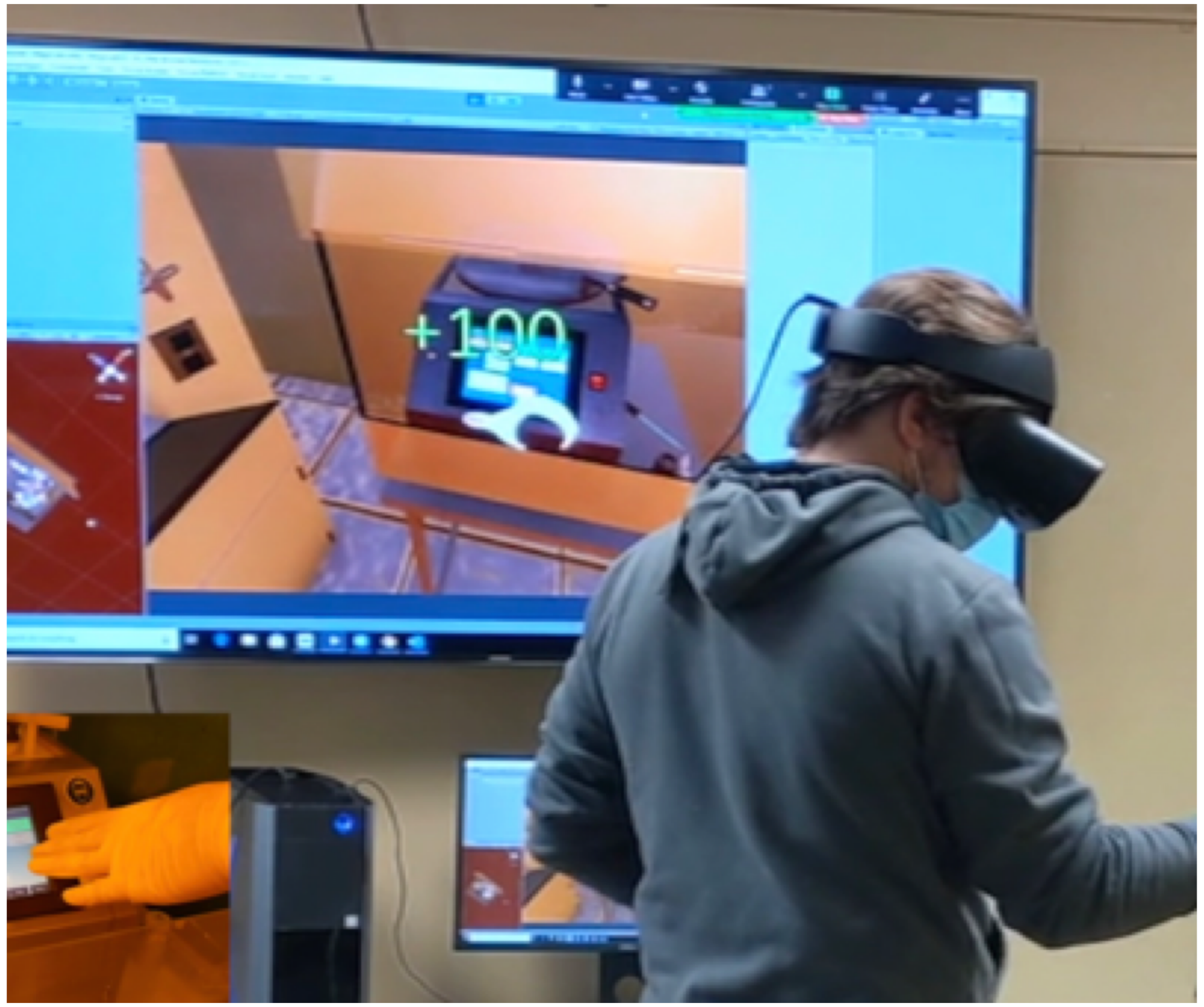
(b)

iVRLab intervention (a) Participants engaged in a training session.

(b) Individual spatial setup.

iVRLab Project Overview

- VR can simulate cleanroom experiences without cost/risk
- Immersive learning boosts engagement, understanding, and motivation
- Can support informal, scalable, and repeatable microelectronics education



Immersive VR Lab Training

School of Information Science & Learning Technologies
College of Engineering
University of Missouri

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iVRLab Features

- Built using Meta Quest 2 VR hardware
- Simulates photolithography cleanroom operations
- Covers four key stages:
    - Wafer cleaning – solvent flush & nitrogen drying
    - Photoresist application – spin coating & pipetting
    - Prebake – hotplate-based wafer baking
    - Mask alignment & exposure – UV alignment and imaging
- Real-time task scoring and feedback
- Object highlighting, pipette/mask aligner simulation
- Instructional overlays for safe, paced training

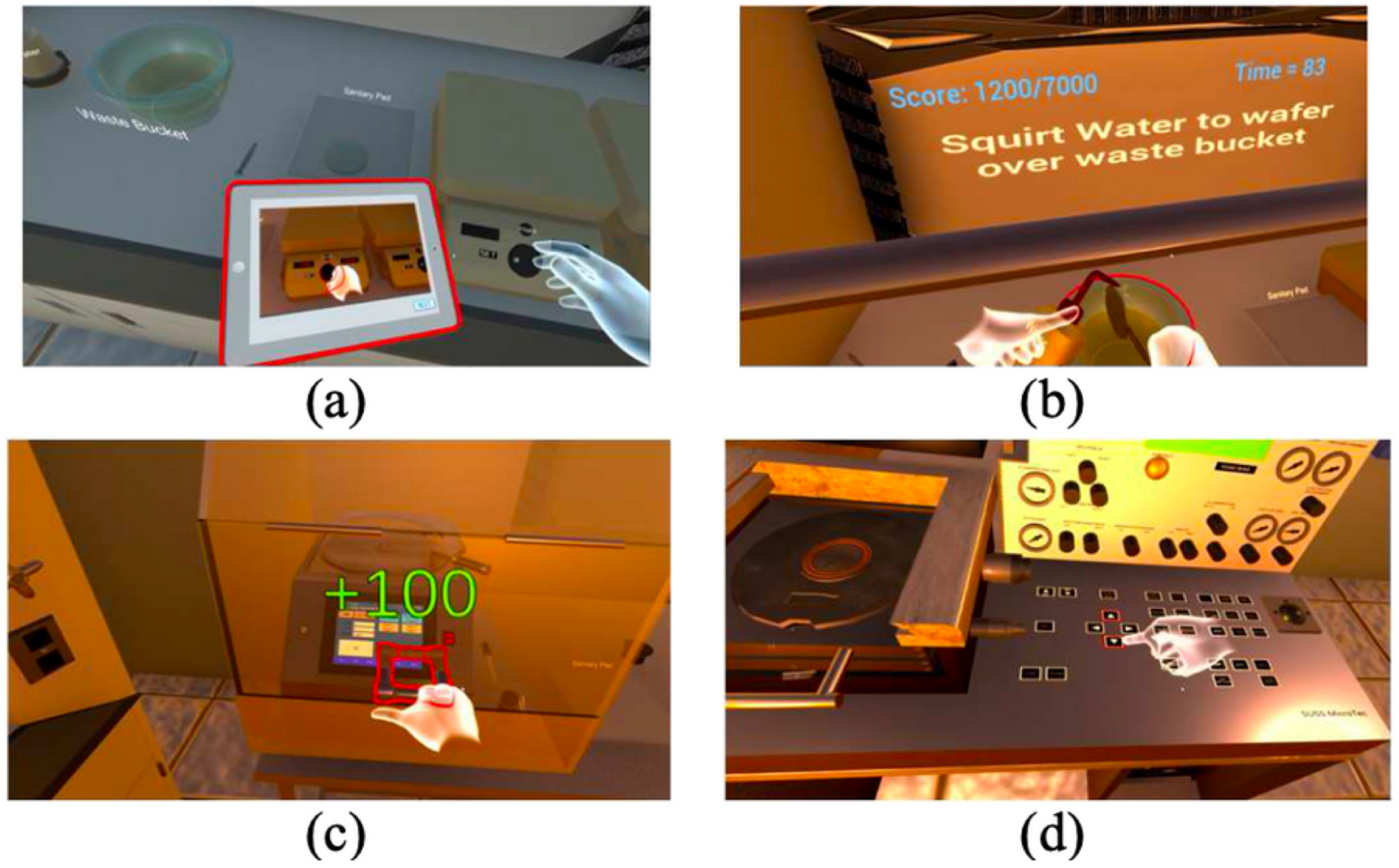


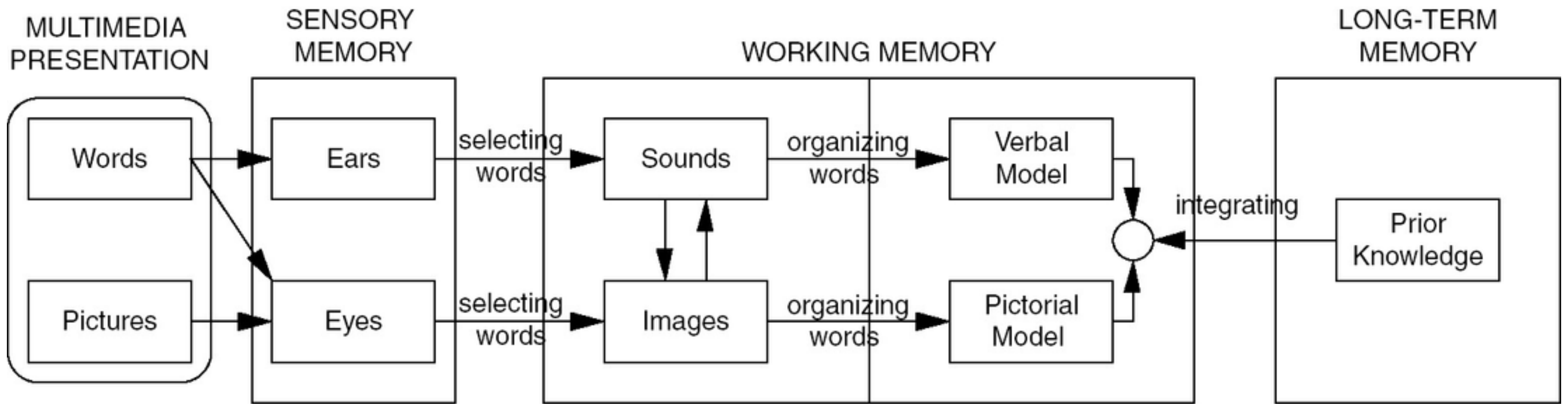
Fig. 1.iVRLab features: (a) Tasks-driven tutorials. (b) Canvas instructions. (c) Points for correct operations. (d) Object highlights.

Research Questions

- RQ1: How does iVRLab affect students' self-efficacy?
- RQ2: How does iVRLab impact students' interest in microelectronics?
- RQ3: How do students perceive their VR learning experience (engagement, immersion, etc.)?

Theoretical Framework

- Based on Mayer's Cognitive Theory of Multimedia Learning (CTML)
- Dual-channel processing, limited cognitive load, active learning
- iVRLab uses scaffolded tasks, interactive feedback, learner-controlled pacing



Mayer RE. Cognitive Theory of Multimedia Learning. In: Mayer RE, ed. The Cambridge Handbook of Multimedia Learning. Cambridge Handbooks in Psychology. Cambridge University Press; 2014:43-71.

Participants

- 27 participants, aged 18–50+
- Educational backgrounds: high school to master's
- Fields: Electrical Engineering, CS, CE, Finance, Industrial Eng.
- Diverse learning needs and goals

Variable	Category	Frequency (n)	Percentage (%)
Gender	Male	19	79.2
	Female	3	12.5
	Non-binary	2	8.3
Age	18-19	4	16.7
	20-23	10	41.7
	24-29	5	20.8
	30-39	4	16.7
	50 and over	1	4.2
Ethnicity	White	11	45.8
	Asian or Asian American	5	20.8
	Black or African American	5	20.8
	Hispanic or Latino	2	8.3
	Middle Eastern	1	4.2
Education	High School Diploma	10	41.7
	Associate Degree	8	33.3
	Bachelor's Degree	4	16.7
	Master's Degree	1	4.2
Field of Study	Electrical Engineering	14	58.3
	Computer Science	5	20.8
	Computer Engineering	3	12.5
	Finance and Economics	1	4.2
	Industrial Engineering	1	4.2

Instruments & Data Collection

Pre/Post Measures

- Self-efficacy – Measured using a modified Motivated Strategies for Learning Questionnaire (MSLQ) [18]
- Interest in Microelectronics & Careers – Adapted from the SITS survey [19]

Post-Only Measures

- Engagement – Instrument adapted from prior successful VR learning research [22]
- Embodiment – Based on embodied self-perception scales [23], [24]
- Immersion – Measured using game immersion scale [25]
- Cognitive Load – Assessed via NASA-TLX and the Virtual Experience Test [20], [21]

Instrument Reliability

- All instruments adapted for microelectronics learning context
- Cronbach's Alpha range: 0.788 – 0.945

Findings – iVRLab's Impact on Self-Efficacy and Interest

Self-Efficacy (Statistically Significant Improvement)

- A paired-samples t-test showed a significant increase in self-efficacy following the intervention, (t(N−1) = −2.97, p = 0.0066.)
- Interpretation: Significant improvement in student confidence with cleanroom procedures

Construct	Pre		Post	
	M	*SD*	*M*	*SD*
Self-efficacy	4.11	0.60	4.37	0.56

Note. Ratings were made on a 5-point Likert scale.

Interest in Microelectronics (Sustained High Baseline)

- Paired-samples t-tests revealed slight increases for ideas about careers (t = −0.89, p = 0.38) and ideas about microelectronics (t = −0.72, p = 0.48).
- Interpretation: Slight, non-significant increases likely due to ceiling effect

Construct	Pre		Post	
	M	*SD*	*M*	*SD*
Ideas about Careers	3.59	0.36	3.64	0.39
Ideas about Microelectronics	3.68	0.43	3.74	0.36

Note. Ratings were made on a 4-point Likert scale.

Key Findings – Learning Experience

Engagement was:
- Strongly correlated with Embodiment (*$r = 0.91, p < 0.001$*)
- Moderately correlated with Immersion (*$r = 0.60, p < 0.01$*)

Embodiment also correlated with Immersion (*$r = 0.66, p < 0.001$*)

Cognitive Load showed no significant correlation with any other variable
- Engagement (*$r = -0.03, p = 0.87$*)
- Embodiment (*$r = -0.15, p = 0.47$*)
- Immersion (*$r = 0.08, p = 0.69$*)
  - Quote: “You can learn *more during* VR *than going in the lab.*”

Variables	1	2	3	4
1. Cognitive Load	1.00			
2. Engagement	-0.03	1.00		
3. Embodiment	-0.15	0.91***	1.00	
4. Immersion	0.08	0.60**	0.66***	1.00

Note: ***$p < 0.001$, **$p < 0.01$, $p < 0.05$

Cognitive Load – Descriptive Analysis

Item	*M*	*SD*
I felt the task very much mentally demanding	2.96	1.28
I found the task very much physically demanding	2.58	1.21
I found I am very much rushed in doing the task	2.62	1.20
I successfully accomplished what I was asked for	3.92	0.80
I worked very hard to accomplish my level of performance	4.15	0.92
I felt insecure and discouraged while doing the task	1.38	0.75
I felt irritated, stressed, and annoyed while doing the task	1.85	1.12

- Participants worked hard and performed well, with high scores on effort and success.
- Low emotional distress: Insecurity and frustration levels were low.
- Cognitive demands were present, but the task was not overwhelming.
- The iVRLab provided a challenging yet supportive learning environment.

Implications and Limitations

Implications for Education
- VR bridges theory and practice in microelectronics
- Enables safe, repeatable, skill-based learning
- Reduces lab demands; supports broader access and inclusion
- Integrates well into informal and formal STEM education

Limitations
- Small sample size and short session time
- Focused only on photolithography
- No control group for direct comparison
- Some technical issues (e.g., headset discomfort)

Conclusion & Future Directions

Conclusion

- iVRLab enhances self-efficacy and maintains STEM interest
- Immersive VR supports active, embodied learning
- Promising for scalable STEM education
- Continued refinement will help meet workforce demands

Future Directions

- Expand to larger and more diverse cohorts
- Longitudinal studies to assess impact over time
- Extend VR to other domains (e.g., circuit design)
- Compare VR with traditional lab training
- Study outcomes in underrepresented learner populations

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Contact:

- Xuxinhao@misouri.edu